Import Libraries

In [ ]:
```
import pandas as pd
import numpy as np
```

Data Analysis from API set

In [ ]:
```
df = pd.read_csv('/Users/frederickhuang/Documents/VSCode/PY/SpaceX Rocket Launches/Falcon9_api_data.csv')
df.head()
```

Out[ ]:

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
0	1	2010-06-04	Falcon 9	6123.547647	LEO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857

In [ ]:
```
# Identify and calculate the percentage of the missing values in each attribute
df.isnull().sum()/len(df)*100
```

Out[ ]:
```
FlightNumber       0.000000
Date               0.000000
BoosterVersion     0.000000
PayloadMass        0.000000
Orbit              0.000000
LaunchSite         0.000000
Outcome            0.000000
Flights            0.000000
GridFins           0.000000
Reused             0.000000
Legs               0.000000
LandingPad        28.888889
Block              0.000000
ReusedCount        0.000000
Serial             0.000000
Longitude          0.000000
Latitude           0.000000
dtype: float64
```

In [ ]:
```
# Identify which columns are numerical and categorical:
df.dtypes
```

Out[ ]:
```
FlightNumber        int64
Date               object
BoosterVersion     object
PayloadMass       float64
Orbit              object
LaunchSite         object
Outcome            object
Flights             int64
GridFins             bool
Reused               bool
Legs                 bool
LandingPad         object
Block             float64
ReusedCount         int64
Serial             object
Longitude         float64
Latitude          float64
dtype: object
```

Calculate the number of launches on each site

The data contains several Space X launch facilities: Cape Canaveral Space Launch Complex 40 VAFB SLC 4E , Vandenberg Air Force Base Space Launch Complex 4E (SLC-4E), Kennedy Space Center Launch Complex 39A KSC LC 39A.

In [ ]:
```
df['LaunchSite'].value_counts()
```

Out[ ]:
```
LaunchSite
CCSFS SLC 40    55
KSC LC 39A      22
VAFB SLC 4E     13
Name: count, dtype: int64
```

Launch Types

LEO: Low Earth orbit (LEO)is an Earth-centred orbit with an altitude of 2,000 km (1,200 mi) or less (approximately one-third of the radius of Earth), or with at least 11.25 periods per day (an orbital period of 128 minutes or less) and an eccentricity less than 0.25. Most of the manmade objects in outer space are in LEO.

VLEO: Very Low Earth Orbits (VLEO) can be defined as the orbits with a mean altitude below 450 km. Operating in these orbits can provide a number of benefits to Earth observation spacecraft as the spacecraft operates closer to the observation.

GTO(Geostationary Transfer Orbit): A geostationary transfer orbit is an elliptical Earth orbit used to transfer satellites from low Earth orbit (LEO) to geostationary orbit (GEO). In a GTO, the perigee (closest point to Earth) is much lower than GEO altitude, while the apogee (farthest point) reaches approximately 22,236 miles (35,786 kilometers) above Earth's equator, the altitude of a geostationary orbit. Satellites in GTO use onboard propulsion to circularize their orbit at GEO altitude, where they can provide services such as weather monitoring, communications, and surveillance.

SSO (or SO): It is a Sun-synchronous orbit also called a heliosynchronous orbit is a nearly polar orbit around a planet, in which the satellite passes over any given point of the planet's surface at the same local mean solar time.

ES-L1 :At the Lagrange points the gravitational forces of the two large bodies cancel out in such a way that a small object placed in orbit there is in equilibrium relative to the center of mass of the large bodies. L1 is one such point between the sun and the earth.

HEO A highly elliptical orbit, is an elliptic orbit with high eccentricity, usually referring to one around Earth.

ISS A modular space station (habitable artificial satellite) in low Earth orbit. It is a multinational collaborative project between five participating space agencies: NASA (United States), Roscosmos (Russia), JAXA (Japan), ESA (Europe), and CSA (Canada)

MEO Geocentric orbits ranging in altitude from 2,000 km (1,200 mi) to just below geosynchronous orbit at 35,786 kilometers (22,236 mi). Also known as an intermediate circular orbit. These are "most commonly at 20,200 kilometers (12,600 mi), or 20,650 kilometers (12,830 mi), with an orbital period of 12 hours

HEO Geocentric orbits above the altitude of geosynchronous orbit (35,786 km or 22,236 mi)

GEO It is a circular geosynchronous orbit 35,786 kilometres (22,236 miles) above Earth's equator and following the direction of Earth's rotation

PO It is one type of satellites in which a satellite passes above or nearly above both poles of the body being orbited (usually a planet such as the Earth)

In [ ]:
```
# Find the number and occurrence of each orbit
df.Orbit.value_counts()
```

Out[ ]:
```
Orbit
GTO      27
ISS      21
VLEO     14
PO        9
LEO       7
SSO       5
MEO       3
ES-L1     1
HEO       1
SO        1
GEO       1
Name: count, dtype: int64
```

In [ ]:
```
# Find the number and occurence of mission outcome of the orbits
landing_outcomes = df['Outcome'].value_counts()

for i,outcome in enumerate(landing_outcomes.keys()):
    print(i,outcome)
```

```
0 True ASDS
1 None None
2 True RTLS
3 False ASDS
4 True Ocean
5 False Ocean
6 None ASDS
7 False RTLS
```

True Ocean means the mission outcome was successfully landed to a specific region of the ocean while False Ocean means the mission outcome was unsuccessfully landed to a specific region of the ocean.

True RTLS means the mission outcome was successfully landed to a ground pad False RTLS means the mission outcome was unsuccessfully landed to a ground pad.

True ASDS means the mission outcome was successfully landed to a drone ship False ASDS means the mission outcome was unsuccessfully landed to a drone ship.

None ASDS and None None these represent a failure to land.

Create Labels for Landing Outcomes

In [ ]:
```
bad_outcomes=set(landing_outcomes.keys()[[1,3,5,6,7]])
bad_outcomes
```

Out[ ]:
```
{'False ASDS', 'False Ocean', 'False RTLS', 'None ASDS', 'None None'}
```

In [ ]:
```
# We want bad landing to be '0', '1' otherwise
landing_class = []
for i in df['Outcome']:
    if i in bad_outcomes:
        landing_class.append(0)
    else:
        landing_class.append(1)
```

In [ ]:
```
# Create new column in the dataframe with outcome label corresponding to each flight.
df['Class']=landing_class
df[['Class']].head(8)
```

Out[ ]:

	Class
0	0
1	0
2	0
3	0
4	0
5	0
6	1
7	1

In [ ]:
```
# Success rate of each flight
df["Class"].mean()
```

Out[ ]:
```
np.float64(0.6666666666666666)
```

Export Labeled Data

In [ ]:
```
# df.to_csv("api_data_with_outcome_labels.csv", index=False)
```

In [ ]: